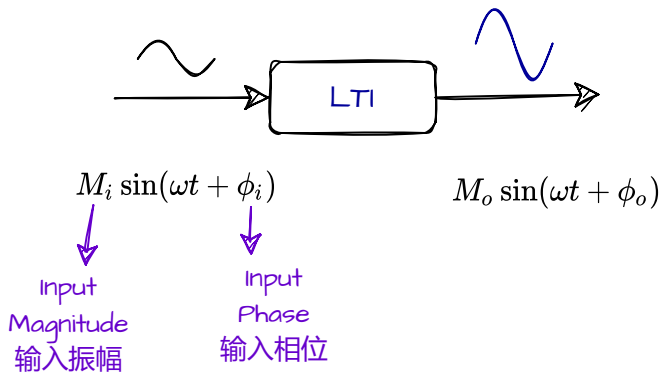


动态系统的建模与分析 8

频率响应与滤波器

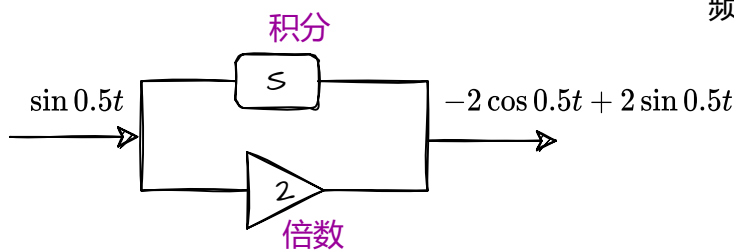
控制和信号系统至关重要

$G(j\omega)$ 玄学? 数学!

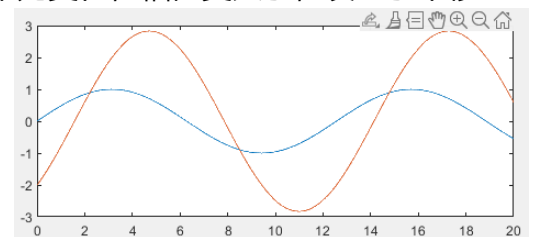


$$\frac{M_o}{M_I} = M \quad \text{Magnitude Response 振幅响应}$$

$$\phi_o - \phi_i = \phi \quad \text{Phase Response 幅角响应}$$



频率无变化, 幅值变大了, 发生了平移



```
t=[0:0.02:20];
f1=sin(0.5*t);
f2=-2*cos(0.5*t)+2*sin(0.5*t);
plot(t,f1);hold on;
plot(t,f2);
```

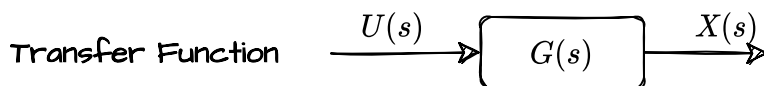
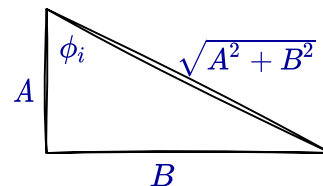
$$u(t) = A \sin \omega t + B \cos \omega t$$

$$= \sqrt{A^2 + B^2} \left(\frac{A}{\sqrt{A^2 + B^2}} \sin \omega t + \frac{B}{\sqrt{A^2 + B^2}} \cos \omega t \right)$$

$$= \sqrt{A^2 + B^2} (\cos \phi_i \sin \omega t + \sin \phi_i \cos \omega t)$$

$$= \sqrt{A^2 + B^2} (\sin(\omega t + \phi_i)) = M_i \sin(\omega t + \phi_i)$$

M_i



$$X(s) = U(s)G(s)$$

$$U(s) = \mathcal{L}[U(t)] = \frac{A\omega}{s^2 + \omega^2} + \frac{Bs}{s^2 + \omega^2} = \frac{A\omega + Bs}{s^2 + \omega^2} = \frac{A\omega + Bs}{(s + j\omega)(s - j\omega)} \quad j = \sqrt{-1}$$

$$G(s) = \frac{N(s)}{D(s)} = \frac{N(s)}{(s - P_1)(s - P_2) \cdots (s - P_n)}$$

$P_1, P_2 \cdots P_n$ Poles 极点

$$X(s) = U(s)G(s) = \frac{A\omega + Bs}{s^2 + \omega^2} \cdot \frac{N(s)}{(s - P_1)(s - P_2) \cdots (s - P_n)}$$

待定系数法

$$= \frac{K_1}{s + j\omega} + \frac{K_2}{s - j\omega} + \frac{C_1}{s - P_1} + \frac{C_2}{s - P_2} + \cdots \frac{C_n}{s - P_n}$$

$(A\omega + Bs)N(s)$

$$= \frac{K_1(s - j\omega)D(s) + K_2(s + j\omega)D(s) + C_1(s - j\omega)(s + j\omega)(\cdots) + C_2(s + j\omega)(s - j\omega)(\cdots) + \cdots}{(s - j\omega)(s + j\omega)(s - P_1)(s - P_2) \cdots (s - P_n)}$$

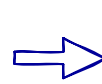
$$X(t) = \mathcal{L}^{-1}(X(s)) = K_1 e^{-j\omega t} + K_2 e^{j\omega t} + C_1 e^{P_1 t} + C_2 e^{P_2 t} + \cdots C_n e^{P_n t}$$

对于稳定系统:

$P_1, P_2, \dots$ 的实部小于0, 例如

$$P_1 = -2$$

$$P_2 = -1 + i$$



$$t \rightarrow \infty$$

$$C_1 e^{-2t} + C_2 e^{-1+it} = 0$$

$$X_{ss}(t) = K_1 e^{-j\omega t} + K_2 e^{j\omega t}$$

steady state

稳态

$$K_1(s - j\omega)D(s) + K_2(s + j\omega)D(s) + C_1(s - j\omega)(s + j\omega)(\dots) \dots = (A\omega + Bs)N(s)$$

let $s = -j\omega$

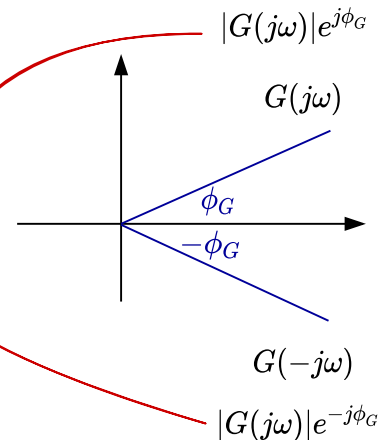
$$K_1(-j\omega - j\omega)D(-j\omega) + 0 + 0 \dots + 0 = (A\omega - Bj\omega)N(-j\omega)$$

$$K_1 = \frac{A\omega - Bj\omega}{-2j\omega} = \frac{N(-j\omega)}{D(-j\omega)}$$

let $s = j\omega$

$$K_1 = \frac{B + Aj}{2} G(-j\omega)$$

$$K_2 = \frac{B - Aj}{2} G(j\omega)$$



$$\begin{aligned} X_{ss}(t) &= \frac{B + Aj}{2} |G(j\omega)| e^{j\phi_G} e^{-j\omega t} + \frac{B - Aj}{2} |G(j\omega)| e^{j\phi_G} e^{j\omega t} \\ &= \frac{1}{2} |G(j\omega)| ((B + Aj)e^{-(\phi_G + \omega t)j} + (B - Aj)e^{(\phi_G + \omega t)j}) \end{aligned}$$

欧拉公式

$$e^{j\theta} = \cos \theta + j \sin \theta$$

$$e^{-(\phi_G + \omega t)j} = \cos(-(\phi_G + \omega t)) + j \sin(-(\phi_G + \omega t))$$

$$= \cos(\phi_G + \omega t) - j \sin(\phi_G + \omega t)$$

$$e^{(\phi_G + \omega t)j} = \cos((\phi_G + \omega t)) + j \sin(\phi_G + \omega t)$$

$$\begin{aligned} X_{ss}(t) &= \frac{1}{2} |G(j\omega)| (B \cos(\phi_G + \omega t) - Bj \sin(\phi_G + \omega t) + Aj \cos(\phi_G + \omega t) + A \sin(\phi_G + \omega t) \\ &\quad + B \cos(\phi_G + \omega t) + Bj \sin(\phi_G + \omega t) - Aj \cos(\phi_G + \omega t) + A \sin(\phi_G + \omega t)) \end{aligned}$$

$$\begin{aligned} X_{ss}(t) &= \frac{1}{2} |G(j\omega)| (2B \cos(\phi_G + \omega t) + 2A \sin(\phi_G + \omega t)) \\ &= |G(j\omega)| \sqrt{A^2 + B^2} \left(\frac{A}{\sqrt{A^2 + B^2}} \cos(\phi_G + \omega t) + \frac{A}{\sqrt{A^2 + B^2}} \sin(\phi_G + \omega t) \right) \end{aligned}$$

$$= |G(j\omega)| M_i \sin(\omega t + \phi_i + \phi_G)$$

$$M_G = |G(j\omega)|$$

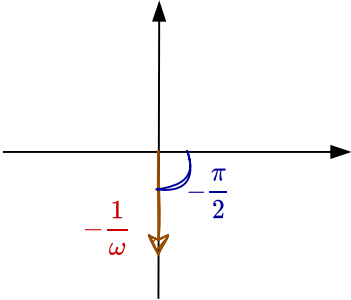
$$\phi_G = \angle G(j\omega)$$

积分例子

$$\int U(t)dt$$

$$G(s) = \frac{1}{s}$$

$$G(j\omega) = \frac{1}{j\omega} = -\frac{1}{\omega}j$$

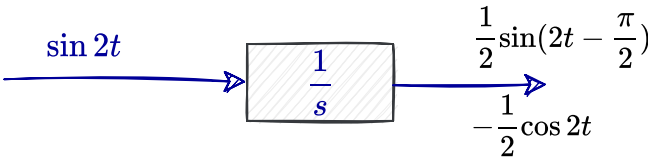
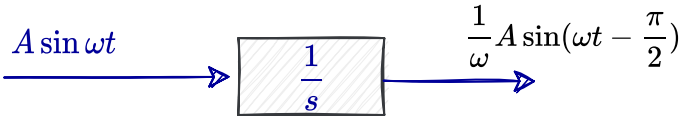


$$|G(j\omega)| = \frac{1}{\omega}$$

$$\omega \uparrow$$

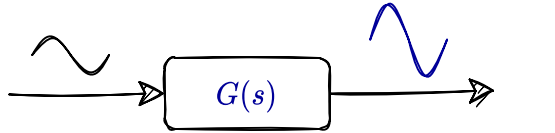
$$|G(j\omega)| \downarrow$$

低通



动态系统的建模与分析 9

一阶系统的频率响应—低通滤波器



$M_i \sin(\omega t + \phi_i)$
Input
Magnitude
输入振幅

$M_o \sin(\omega t + \phi_o)$
Input
Phase
输入相位

$$\frac{M_o}{M_i} = |G(j\omega)|$$

$$\phi_o - \phi_i = \angle G(j\omega)$$

Magnitude Response
振幅响应

Phase Response
幅角响应

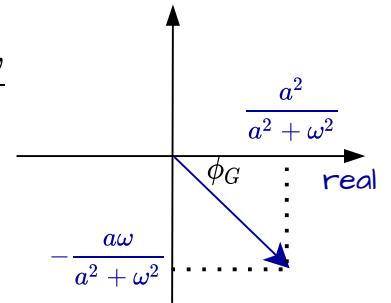
1st Order System
一阶系统

$$G(s) = \frac{a}{s + a}$$

$$s = j\omega \Rightarrow G(j\omega) = \frac{a}{a + j\omega} = \frac{a(a - j\omega)}{(a + j\omega)(a - j\omega)} = \frac{a^2 - aj\omega}{a^2 + \omega^2}$$

$$= \frac{a^2}{a^2 + \omega^2} + (-\frac{a\omega}{a^2 + \omega^2})j$$

Real Imaginary



$$|G(j\omega)| = \sqrt{\left(\frac{a^2}{a^2 + \omega^2}\right)^2 + \left(\frac{a\omega}{a^2 + \omega^2}\right)^2}$$

$$= \sqrt{\frac{a^2(a^2 + \omega^2)}{a^2 + \omega^2}}$$

$$= \sqrt{\frac{1}{1 + \left(\frac{\omega}{a}\right)^2}}$$

$$\omega \ll a \Rightarrow \left(\frac{\omega}{a}\right)^2 \rightarrow 0 \Rightarrow |G(j\omega)| \rightarrow 1$$

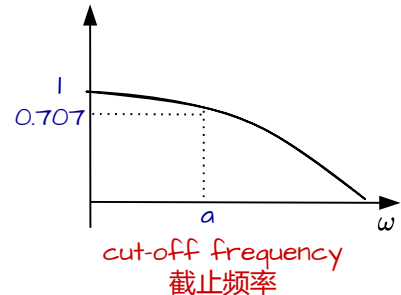
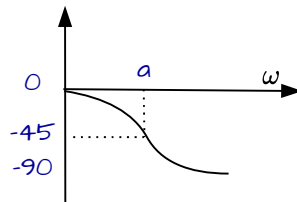
$$\omega \uparrow \Rightarrow \left(\frac{\omega}{a}\right)^2 \uparrow \Rightarrow |G(j\omega)| \downarrow$$

$$\omega = a \Rightarrow |G(j\omega)| = \sqrt{\frac{1}{2}} = 0.707$$

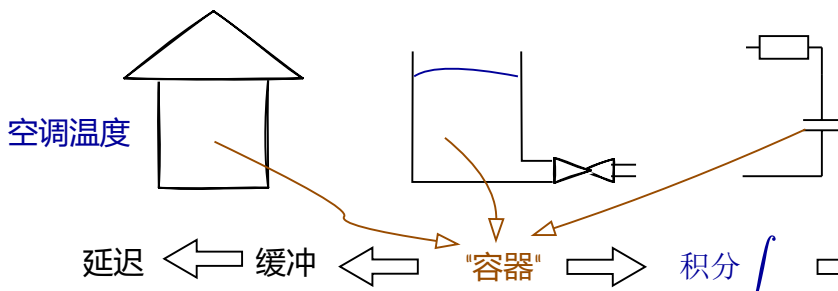
$$\omega \gg a \Rightarrow \left(\frac{\omega}{a}\right)^2 \rightarrow \infty \Rightarrow |G(j\omega)| \rightarrow 0$$

$$\phi_G = \arctan\left(-\frac{a\omega}{a^2}\right)$$

$$= -\arctan\left(\frac{\omega}{a}\right)$$

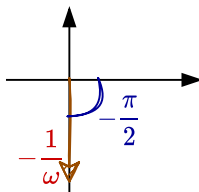


低通滤波器 Low Pass Filter



$$\frac{1}{s} \quad s = j\omega \quad \frac{1}{j\omega} = -\frac{1}{\omega}j$$

$$|G\left(\frac{1}{j\omega}\right)| = \frac{1}{\omega}$$



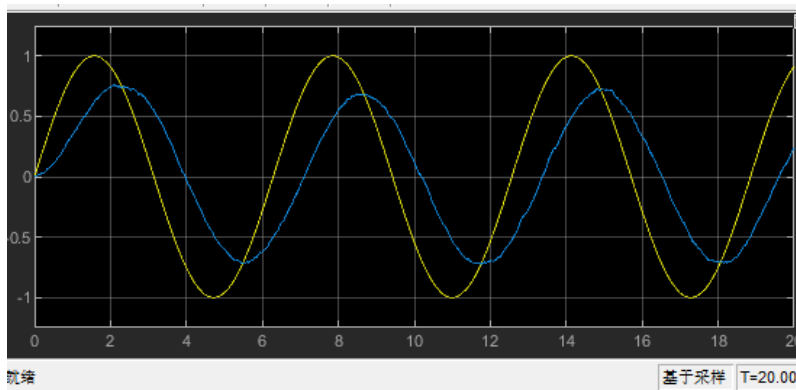
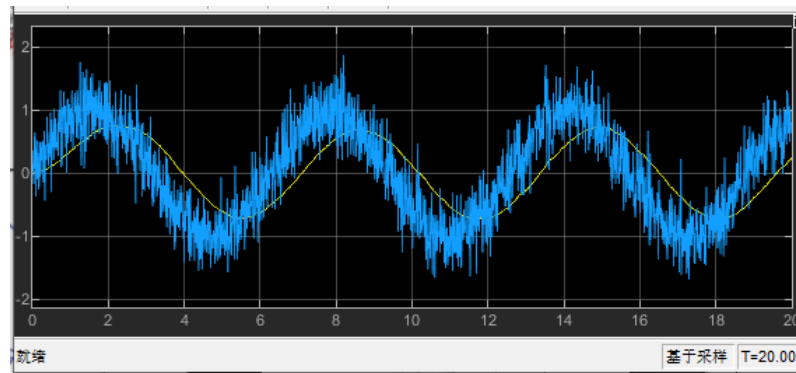
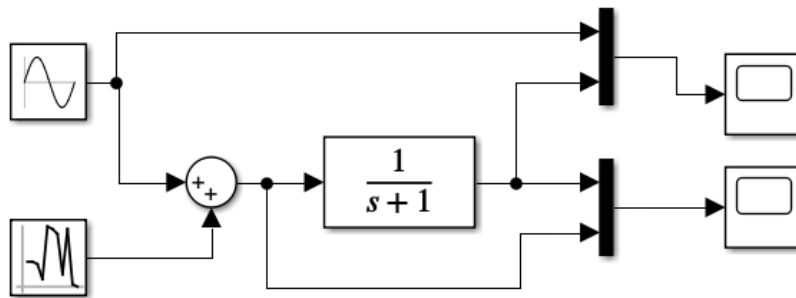
积累 → 曾经沧海 → 处乱不惊

Be Responsibility

放下 → 解放思想 → 逐风追电

做一个负责任的人

Simulink 仿真



动态系统的建模与分析

二阶系统动态响应 2nd order system

Newton's 2nd Law

位移对时间的二次导

$$F = ma$$

a : 加速度

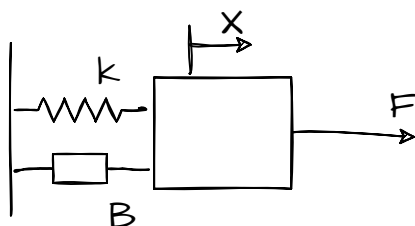
$$\frac{d^2 x}{dt^2}$$

$$\ddot{x}$$

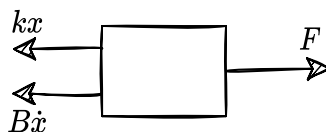
动力学和运动学

Vibration 振动

分析二阶系统 matlab/simulink



mass-spring-damping



定义一些参数

$$m\ddot{x} = F - B\dot{x} - kx$$

$$\ddot{x} + \frac{B}{m}\dot{x} + \frac{k}{m}x = \frac{F}{m}$$

$$\omega_n = \sqrt{\frac{k}{m}}$$

Natural Frequency 固有频率

$$\zeta = \frac{B}{2\sqrt{km}}$$

Damping Ratio 阻尼比

Response to Initial Conditions 初始条件

$$F = 0 \quad x(0) = x_0 \quad \dot{x}(0) = \dot{x}_0$$

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = 0$$

求解微分方程

simulink

$$\ddot{x} = -2\zeta\omega_n\dot{x} - \omega_n^2x$$

$$x(0) = 5$$

$$\dot{x}(0) = 0$$

$$x(t) = e^{\lambda t}$$

$$\dot{x} = \lambda e^{\lambda t}$$

$$\ddot{x} = \lambda^2 e^{\lambda t}$$

$$\lambda^2 e^{\lambda t} + 2\zeta\lambda e^{\lambda t} + \omega_n^2 e^{\lambda t} = 0$$

$$(\lambda^2 + 2\zeta\omega_n\lambda + \omega_n^2)e^{\lambda t} = 0 \quad e^{\lambda t} \neq 0$$

$$\lambda^2 + 2\zeta\omega_n\lambda + \omega_n^2 = 0$$

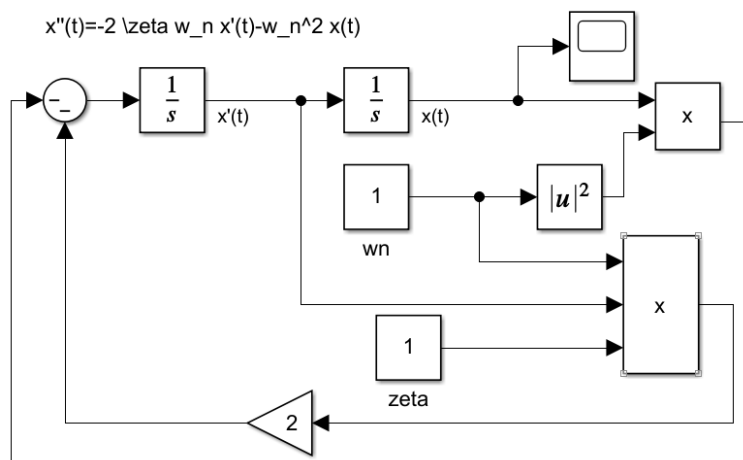
characteristic equation
特征方程

$$\lambda = \frac{-2\zeta\omega_n \pm \sqrt{4\zeta^2\omega_n^2 - 4\omega_n^2}}{2}$$

$$\lambda_1 = -\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1}$$

$$\lambda_2 = -\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}$$

$$x''(t) = -2\zeta\omega_n x'(t) - \omega_n^2 x(t)$$

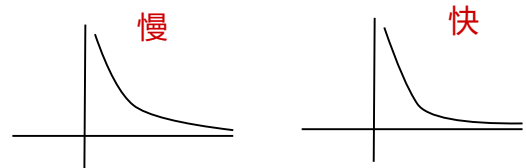


分情况讨论

① $\zeta > 1$ $\frac{B}{2\sqrt{km}} > 1$ over damped 过阻尼

$$\lambda_1 = -\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1} < 0$$

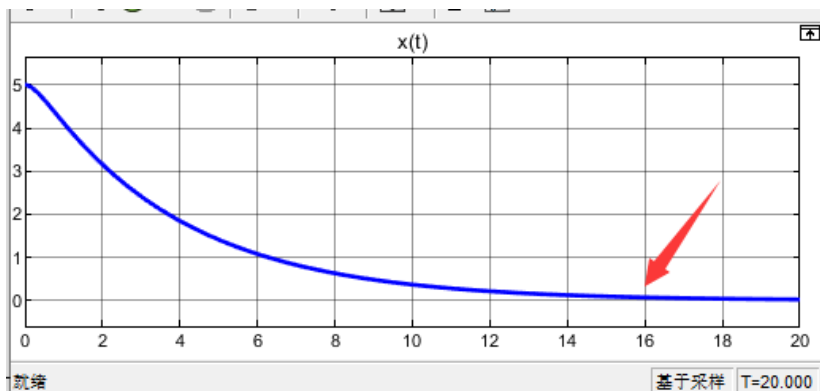
$$\lambda_2 = -\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1} < 0$$



$$x(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$$

$|\lambda_1| < |\lambda_2|$

simulink中
 $\zeta = 2$



② $\zeta = 1$ $\frac{B}{2\sqrt{km}} = 1$ critical damped 临界阻尼

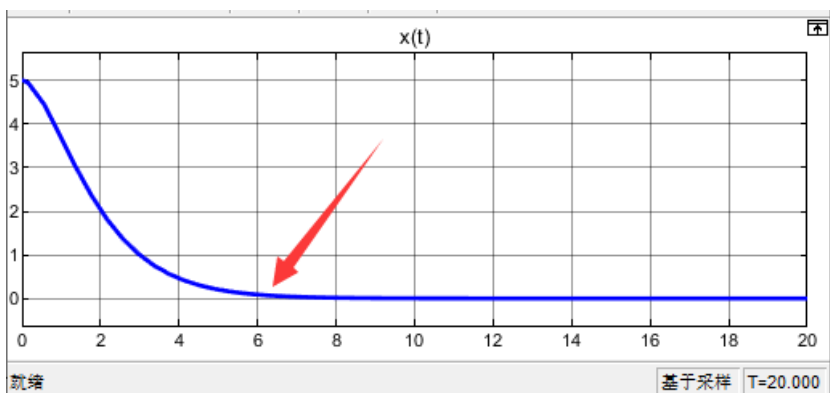
$$\lambda_1 = \lambda_2 = -\omega_n$$

$$x(t) = (C_1 + C_2 t) e^{\lambda t}$$

不存在一个慢的收敛项，
收敛比过阻尼快

$$|\lambda_1| < \lambda < |\lambda_2|$$

simulink中
 $\zeta = 1$



③ $0 < \zeta < 1$ $0 < \frac{B}{2\sqrt{km}} < 1$

underdamped 欠阻尼

ω_d Damped natural frequency
阻尼固有频率

$$\lambda = -\zeta\omega_n \pm i\omega_n\sqrt{1 - \zeta^2}$$

$$x(t) = e^{-\zeta\omega_n t} (C_1 \cos \omega_d t + C_2 \sin \omega_d t)$$

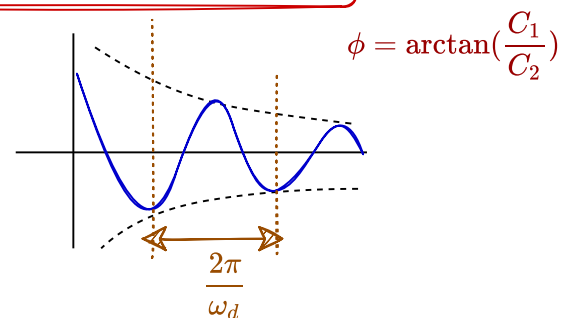
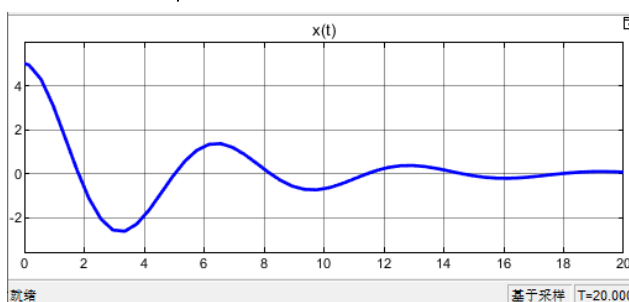
共轭

$$\lambda_1 = -\zeta\omega_n + i\omega_n\sqrt{1 - \zeta^2}$$

$$\lambda_2 = -\zeta\omega_n - i\omega_n\sqrt{1 - \zeta^2}$$

$$x(t) = \sqrt{C_1^2 + C_2^2} e^{-\zeta\omega_n t} (\sin(\omega_d t + \phi))$$

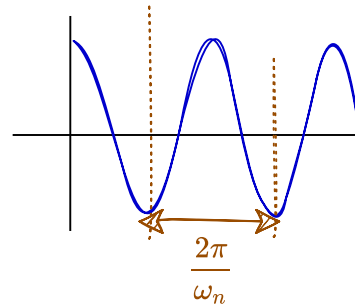
simulink中
 $\zeta = 0.2$



④ $\zeta = 0 \quad B = 0$

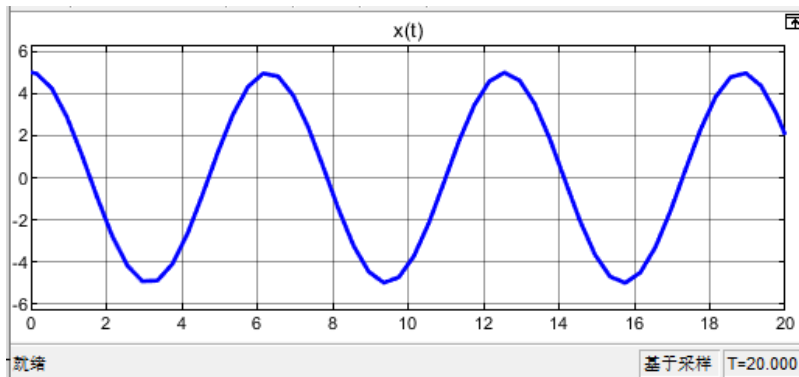
$$x(t) = e^0 (C_1 \cos \omega_n t + C_2 \sin \omega_n t)$$

$$x(t) = \sqrt{C_1^2 + C_2^2} \sin(\omega_d t + \phi)$$

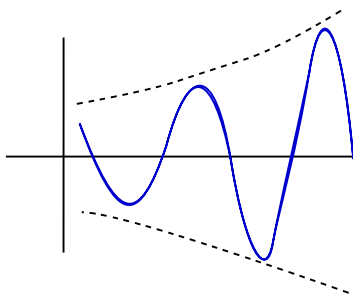


simulink中

$$\zeta = 0$$



⑤ $-1 < \zeta < 0$



不稳定

$$\lambda = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1} < 0$$

$$\lambda_1 = -\zeta\omega_n + i\omega_n \sqrt{1 - \zeta^2}$$

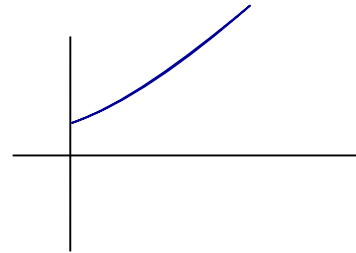
$$\lambda_2 = -\zeta\omega_n - i\omega_n \sqrt{1 - \zeta^2}$$

$$x(t) = e^{-\zeta\omega_n t} (C_1 \cos \omega_n \sqrt{1 - \zeta^2} t + C_2 \sin \omega_n \sqrt{1 - \zeta^2} t)$$

$$x(t) = \sqrt{C_1^2 + C_2^2} e^{-\zeta\omega_n t} (\sin(\omega_d t + \phi))$$

$-\zeta\omega_n > 0$

⑥ $\zeta < -1$



$$\lambda = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1} < \zeta$$

$$\lambda_1 = -\zeta\omega_n + \omega_n \sqrt{\zeta^2 - 1} > 0$$

$$\lambda_2 = -\zeta\omega_n - \omega_n \sqrt{\zeta^2 - 1} > 0$$

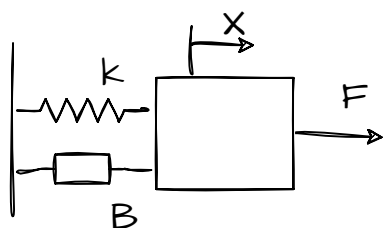
$$x(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$$

> 0

动态系统的建模与分析

2nd order System Unit Step Response 二阶系统单位阶跃

mass-spring-damping



$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = F$$

输入 input $u(t) = \frac{F}{\omega_n^2}$ 单位化

输出 input $x(t)$

$$\zeta = \frac{B}{2\sqrt{km}} \quad \text{Damping Ratio 阻尼比}$$

$$\omega_n = \sqrt{\frac{k}{m}} \quad \text{Natural Frequency 固有频率}$$

o 初始条件

$$x(0) = x_0 \quad \dot{x}(0) = \dot{x}_0$$

$$\mathcal{L}\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = \omega_n^2u(t)$$

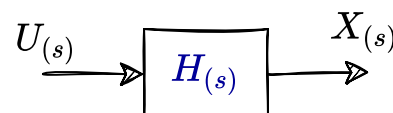
l 微分方程求解

$$x(t) = x_n + x_p$$

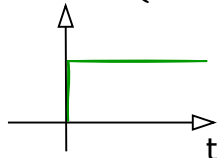
2. 传递函数

$$s^2X(s) + 2\zeta\omega_nsX(s) + \omega_n^2X(s) = \omega_n^2U(s)$$

$$H(s) = \frac{X(s)}{U(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_ns + \omega_n^2}$$

Unit step Input
单位阶跃

$$u(t) = \begin{cases} 0 & t = 0 \\ 1 & t > 0 \end{cases}$$

Find Poles
极点

$$\mathcal{L}[u(t)] = \frac{1}{s}$$

$$X(s) = U(s)H(s)$$

$$X(s) = \frac{1}{s} \frac{\omega_n^2}{s^2 + 2\zeta\omega_ns + \omega_n^2}$$

① 求 $X(s)$ ② $\mathcal{L}^{-1}[X(s)] = x(t)$

$$s^2 + 2\zeta\omega_ns + \omega_n^2 = 0$$

$$s = \frac{-2\zeta\omega_n \pm \sqrt{4\zeta^2\omega_n^2 - 4\omega_n^2}}{2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

under Damped
欠阻尼

$$0 < \zeta < 1$$

$$s = -\zeta\omega_n \pm i\omega_n\sqrt{1 - \zeta^2}$$

$$X(s) = \frac{A}{s - p_1} + \frac{B}{s - p_2} + \frac{C}{s - p_3} = \frac{A(s - p_2)(s - p_3) + B(s - p_1)(s - p_3) + C(s - p_1)(s - p_2)}{(s - p_1)(s - p_2)(s - p_3)} = \omega_n^2$$

$$A(s - p_2)(s - p_3) + B(s - p_1)(s - p_3) + C(s - p_1)(s - p_2) = \omega_n^2$$

$$s = p_1 = 0$$

$$A(-p_2)(-p_3) = \omega_n^2$$

$$A(\zeta\omega_n - i\omega_n\sqrt{1-\zeta^2})(\zeta\omega_n + i\omega_n\sqrt{1-\zeta^2}) = \omega_n^2$$

$$A(\cancel{\zeta^2}\omega_n^2 + \omega_n^2 - \cancel{\omega_n^2}\zeta^2) = \omega_n^2$$

$$A = 1$$

$$s = p_2 = -\zeta\omega_n + i\omega_n\sqrt{1-\zeta^2}$$

$$B(-\zeta\omega_n + i\omega_n\sqrt{1-\zeta^2} - 0)(\cancel{-\zeta\omega_n} + i\omega_n\sqrt{1-\zeta^2} + \cancel{\zeta\omega_n} + i\omega_n\sqrt{1-\zeta^2}) = \omega_n^2$$

$$B(-\zeta\omega_n + i\omega_n\sqrt{1-\zeta^2})(2i\omega_n\sqrt{1-\zeta^2}) = \omega_n^2$$

$$B(-2i\zeta\omega_n^2\sqrt{1-\zeta^2} - 2\omega_n^2(1-\zeta^2)) = \omega_n^2$$

$$B(\cancel{-2\omega_n^2})(i\zeta\sqrt{1-\zeta^2} + (1-\zeta^2)) = \cancel{\omega_n^2}$$

$$\begin{aligned} B &= -\frac{1}{2} \frac{1}{(1-\zeta^2) + i\zeta\sqrt{1-\zeta^2}} \cdot \frac{(1-\zeta^2) - i\zeta\sqrt{1-\zeta^2}}{(1-\zeta^2) - i\zeta\sqrt{1-\zeta^2}} \\ &= -\frac{1}{2} \frac{(1-\zeta^2) - i\zeta\sqrt{1-\zeta^2}}{(1-\zeta^2)^2 + \zeta^2(1-\zeta^2)} = -\frac{1}{2} \frac{(1-\zeta^2) - i\zeta\sqrt{1-\zeta^2}}{1-2\zeta^2 + \cancel{\zeta^4} + \zeta^2 - \cancel{\zeta^4}} \\ &= -\frac{1}{2} \frac{(1-\zeta^2) - i\zeta\sqrt{1-\zeta^2}}{1-\zeta^2} = -\frac{1}{2} \left[1 - \frac{\zeta\sqrt{1-\zeta^2}}{1-\zeta^2} i \right] \end{aligned}$$

$$B = -\frac{1}{2} \left[1 - \frac{\zeta}{\sqrt{1-\zeta^2}} i \right]$$

$$s = p_3 \quad C = B^*$$

$$C = -\frac{1}{2} \left[1 + \frac{\zeta}{\sqrt{1-\zeta^2}} i \right]$$

$$X(s) = \frac{\text{A}}{s} - \frac{\text{B}}{s - p_2} - \frac{\text{C}}{s - p_3}$$

$$\textcircled{2} \mathcal{L}^{-1}[X(s)] = X(t)$$

$$X(t) = 1 \cdot e^{0t} - \frac{1}{2} \left[1 - \frac{\zeta}{\sqrt{1-\zeta^2}} i \right] e^{p_2 t} - \frac{1}{2} \left[1 + \frac{\zeta}{\sqrt{1-\zeta^2}} i \right] e^{p_3 t}$$

Damped natural frequency
阻尼固有频率

$$= 1 - \frac{1}{2} \left[1 - \frac{\zeta}{\sqrt{1-\zeta^2}} i \right] e^{(-\zeta\omega_n + i\omega_n\sqrt{1-\zeta^2})t} - \frac{1}{2} \left[1 + \frac{\zeta}{\sqrt{1-\zeta^2}} i \right] e^{(-\zeta\omega_n - i\omega_n\sqrt{1-\zeta^2})t}$$

ω_d

||

$$= 1 - e^{-\zeta\omega_n t} \left[\frac{1}{2} \left(1 - \frac{\zeta}{\sqrt{1-\zeta^2}} i \right) e^{i\omega_d t} + \frac{1}{2} \left(1 + \frac{\zeta}{\sqrt{1-\zeta^2}} i \right) e^{-i\omega_d t} \right]$$

$$= \cos \omega_d t$$

$$= 1 - e^{-\zeta\omega_n t} \left[\frac{1}{2} (e^{i\omega_d t} + e^{-i\omega_d t}) + \frac{1}{2} \frac{\zeta}{\sqrt{1-\zeta^2}} i (-e^{i\omega_d t} + e^{-i\omega_d t}) \right]$$

$$= \sin \omega_d t$$

$$= 1 - e^{-\zeta\omega_n t} \left[\cos \omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_d t \right]$$

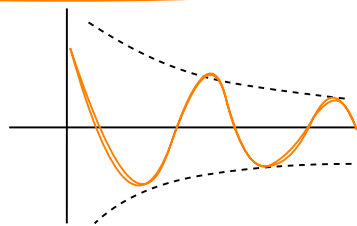
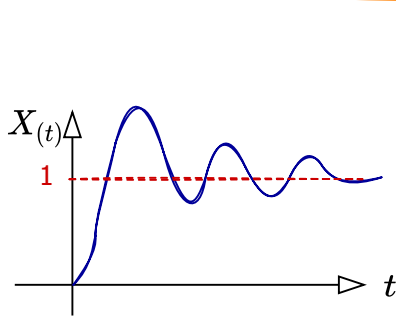
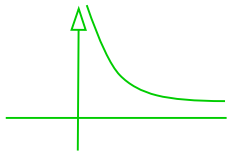
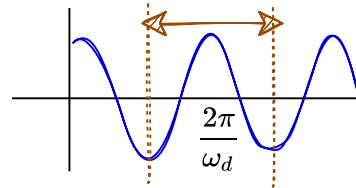
$$= 1 - e^{-\zeta\omega_n t} \sqrt{1 + \frac{\zeta^2}{1-\zeta^2}} (\sin(\omega_d t + \phi))$$

$$A \sin \theta + B \cos \theta$$

$$= \sqrt{A^2 + B^2} \sin(\theta + \phi)$$

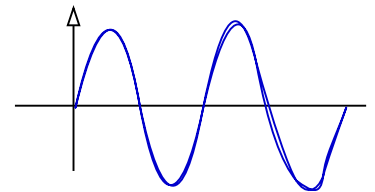
$$\phi = \arctan \frac{B}{A}$$

$$= 1 - e^{-\zeta\omega_n t} \sqrt{\frac{1}{1-\zeta^2}} \sin(\omega_d t + \phi)$$



$\zeta = 0$ no damping

$$X(t) = 1 - \cos \omega_n t$$

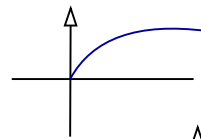


$$t = 0 \quad X(t) = 0$$

$$t \rightarrow \infty \quad X(t) = 1$$

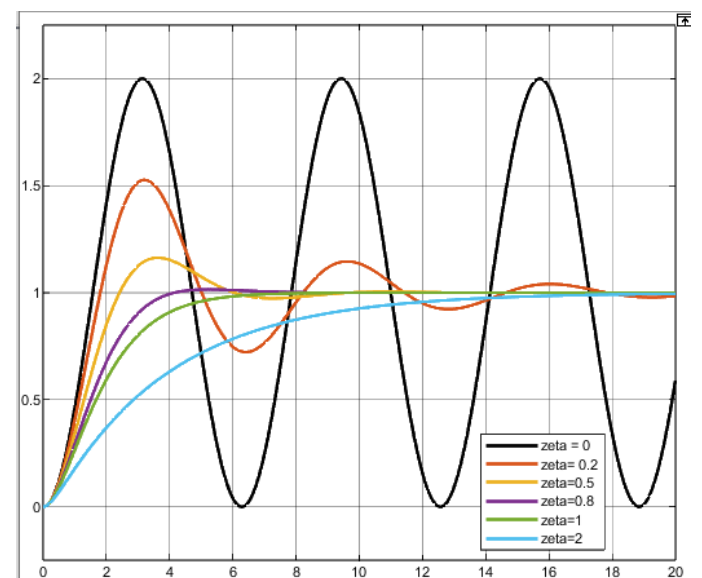
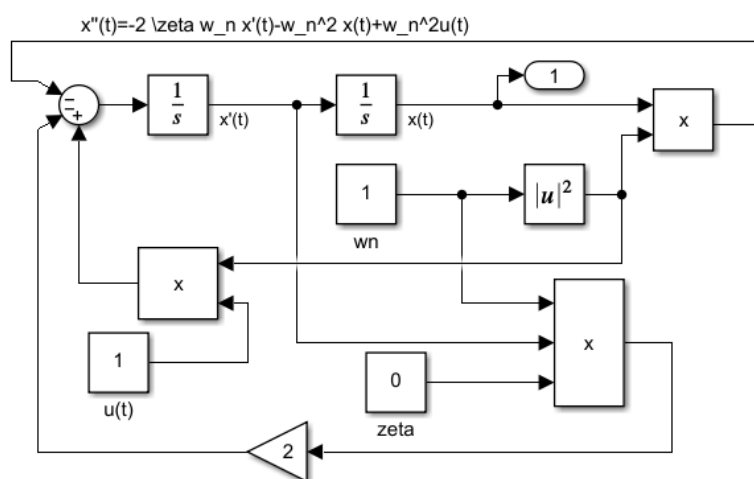
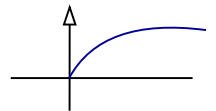
$\zeta = 1$ critical damping

$$X(t) = 1 - e^{-\omega_n t} [1 + \omega_n t]$$



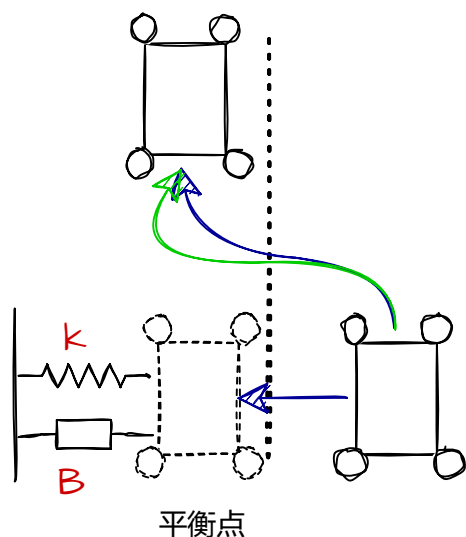
$\zeta > 1$ Over damped

$$X(t) = 1 + \frac{\omega_n}{2\sqrt{\zeta^2 - 1}} e^{(-\zeta + \sqrt{\zeta^2 - 1})\omega_n t} + e^{-\zeta - \sqrt{\zeta^2 - 1}\omega_n t}$$



动态系统的建模与分析

2nd order System Unit Step Response 二阶系统 单位阶跃



平衡点

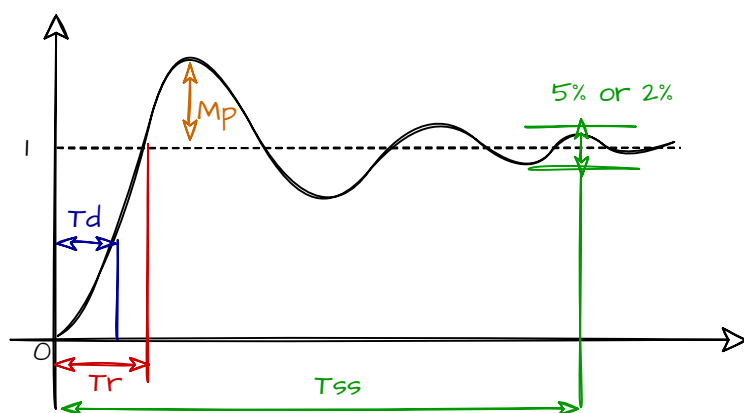
自动驾驶课题: 自动变道

这个二阶系统, 抽象成一个弹簧阻尼系统

如何判断这两个轨迹的算法好呢?

欠阻尼系统为例

$$X(t) = 1 - e^{-\zeta\omega_n t} \sqrt{\frac{1}{1-\zeta^2}} \sin(\omega_d t + \phi)$$



Td: Delay time 延迟时间 50%

Tr: Rise time 上升时间 100%

Mp: Max Overshoot 最大超调量

$$(X_p - 1) \times 100\%$$

Tss: settling time 调节时间, 稳态

$$Tr: \quad 1 = 1 - e^{-\zeta\omega_n t} \sqrt{\frac{1}{1-\zeta^2}} \sin(\omega_d t + \phi) = 0$$

$$\sin(\omega_d t + \phi) = 0$$

$$\omega_d t + \phi = \pi$$

$$tr = \frac{\pi - \phi}{\omega_d}$$

$$Mp: \quad X_{(tp)} \quad \text{peak time} \quad \dot{X}_{(t)} = 0 \quad 1st$$

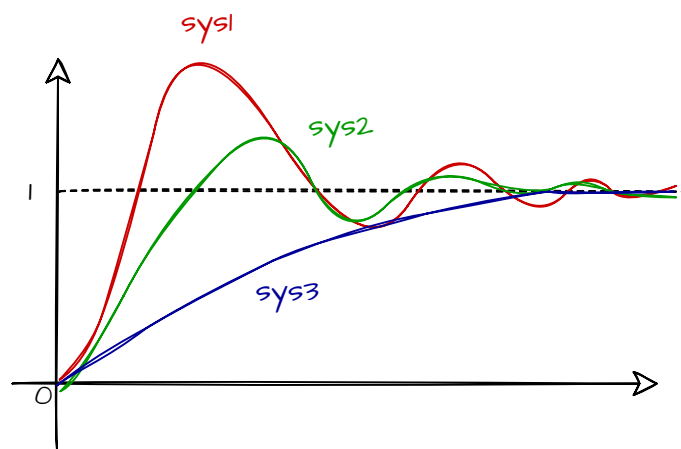
$$tp = \frac{\pi}{\omega_d}$$

$$Mp = e^{-\zeta\pi/\sqrt{1-\zeta^2}} \times 100\%$$

$$Tss: \quad 2\% \quad Tss = \frac{4}{\zeta\omega_n}$$

$$5\% \quad Tss = \frac{3}{\zeta\omega_n}$$

分析



	Tr	Mp	Tss
Sys1	3	1	2
Sys2	2	2	3
Sys3	1	3	1

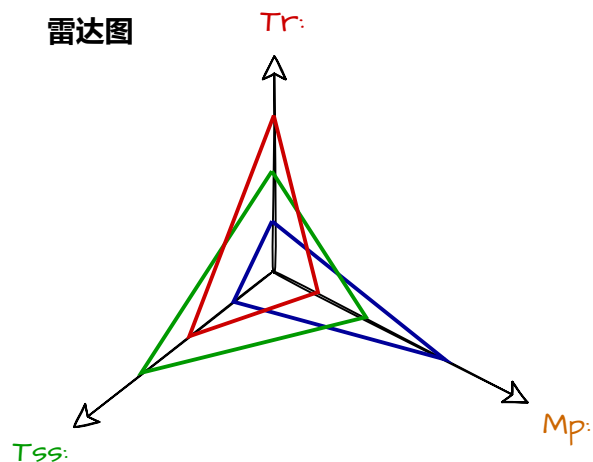
计分规则，1, 2, 3

Tr 越小，分数高

Mp 越小，分数高

Tss 越小，分数高

雷达图



这个圈越大越好，但是要针对不同的系统做出考虑

比如转换车道，我们从超调量入手。
选择3号系统，尽管反应慢一些，但不会驶入其他车道

比如紧急避障，我们应该选择1号系统。
尽快的避开危险区。

当然障碍还远，应该选择2号系统。
既能避开，也不会影响乘车体验。

实际上很多的内容不是对错来判断的。
只是在某一种情况下，哪一种更加适合。

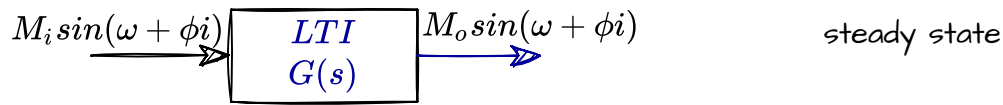
动态系统的建模与分析

2nd order System Frequency Response

二阶系统

频率响应

线性时不变系统



$$M = \frac{M_o}{M_i} = |G(j\omega)|$$

振幅响应

Amplitude response

$$\phi = \phi_o - \phi_i = \angle G(j\omega)$$

幅角响应

Phase response

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

ω_n Natural Frequency 固有频率

ζ Damping Ratio 阻尼比

$$G(j\omega) = \frac{\omega_n^2}{-\omega^2 + 2\zeta\omega_n\omega j + \omega_n^2}$$

$$= \frac{1}{-\frac{\omega^2}{\omega_n^2} + 2\zeta\frac{\omega}{\omega_n}j + 1}$$

输入频率

$$\frac{\omega}{\omega_n} = \Omega$$

$$= \frac{1}{-\Omega^2 + 2\zeta\Omega j + 1} = \frac{1 - \Omega^2 - 2\zeta\Omega j}{(1 - \Omega^2 + 2\zeta\Omega j)(1 - \Omega^2 - 2\zeta\Omega j)}$$

$$= \underbrace{\frac{1 - \Omega^2}{(1 - \Omega^2)^2 + 4\zeta^2\Omega^2}}_{\text{Real}} - \underbrace{\frac{2\zeta\Omega}{(1 - \Omega^2)^2 + 4\zeta^2\Omega^2}j}_{\text{Im}}$$

$$|G(j\omega)| = (Real(G_{(j\omega)})^2 + Im(G_{(j\omega)})^2)^{\frac{1}{2}}$$

$$= \sqrt{\frac{(1 - \Omega^2)^2 + 4\zeta^2\Omega^2}{((1 - \Omega^2)^2 + 4\zeta^2\Omega^2)^2}}$$

$$|G(j\omega)| = \sqrt{\frac{1}{(1 - \Omega^2)^2 + 4\zeta^2\Omega^2}}$$

$$\frac{\omega}{\omega_n} = \Omega$$

$$|G(j\omega)|_{\omega=\omega_n\sqrt{1-2\zeta^2}} = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

分析

$$\Omega = 0 \quad \omega = 0$$

$$|G(j\omega)| = 1$$

$$\Omega \rightarrow \infty \quad \omega \gg \omega_n \quad \omega \rightarrow \infty$$

$$|G(j\omega)| \rightarrow 0$$

$$\Omega = 1 \quad \omega = \omega_n$$

$$|G(j\omega)| = \sqrt{\frac{1}{0 + 4\zeta^2}} = \frac{1}{2\zeta} \quad \begin{matrix} \zeta < 0.5 & |G(j\omega)| > 1 \\ \zeta > 0.5 & |G(j\omega)| < 1 \end{matrix}$$

pole: $f(\Omega) = (1 - \Omega^2)^2 + 4\zeta^2\Omega^2$

$$\frac{df(\Omega)}{d\Omega} = 2(1 - \Omega^2)(-2\Omega) + 8\zeta^2\Omega^2 = 0$$

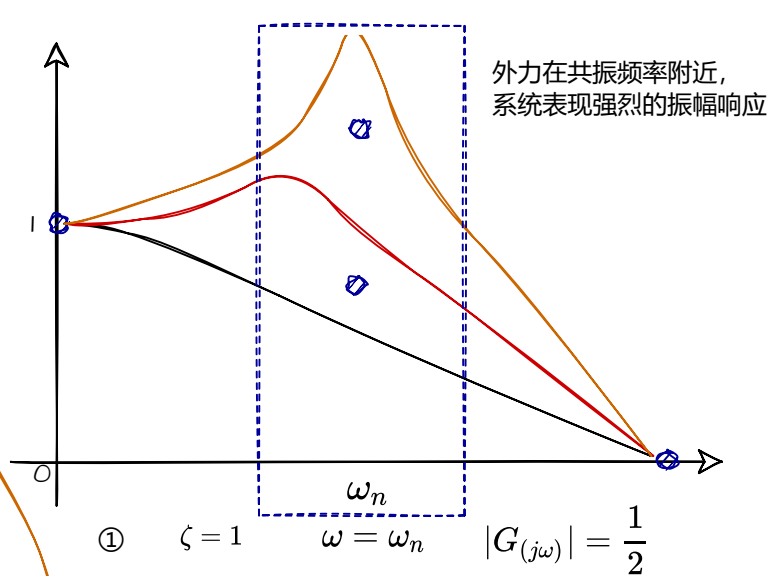
$$\Omega(-1 + \Omega^2 + 2\zeta^2) = 0$$

$$= 0$$

$$\Omega = \sqrt{1 - 2\zeta^2} \quad \Omega = \sqrt{1 - 2\zeta^2} = \frac{\omega}{\omega_n}$$

$$1 - 2\zeta^2 > 0 \text{ 时存在极值} \quad \zeta < \sqrt{\frac{1}{2}}$$

$$|G(j\omega)|_{\omega=\omega_n\sqrt{1-2\zeta^2}} = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$



② $\zeta = 0.5 \quad \omega = \omega_n \quad |G(j\omega)| = 1$

$\omega = \omega_n\sqrt{1 - 2\zeta^2} \quad |G(j\omega)| = 1.16$

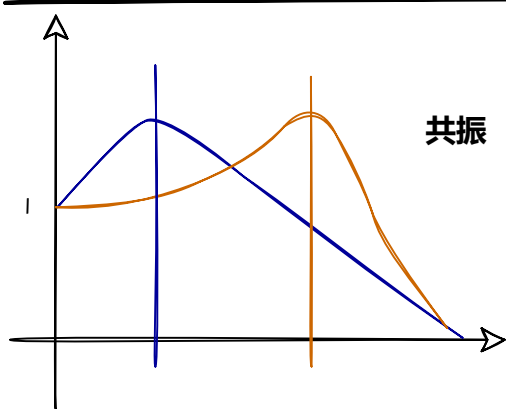
③ $\zeta = 0 \quad \omega = \omega_n \quad |G(j\omega)| = \infty$

$$\omega = \omega_n\sqrt{1 - 2\zeta^2}$$

共振频率

Resonance frequency

$$\zeta \rightarrow 0, \omega \approx \omega_n$$



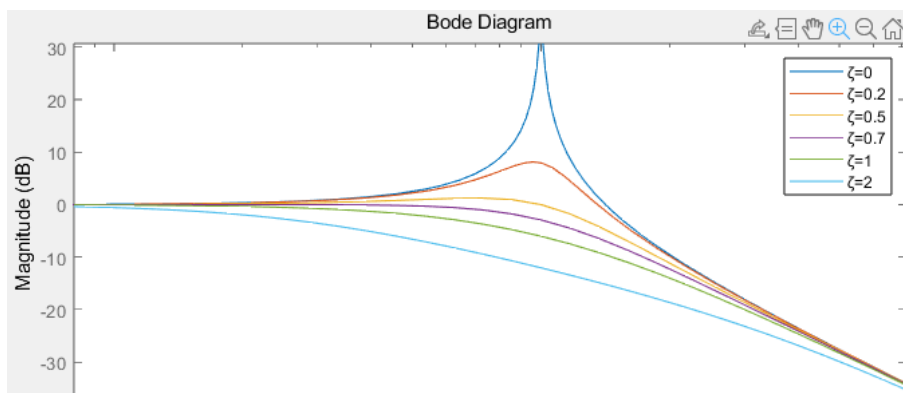
matlab bode函数

生活中：劝人说话，感同身受是很难的事情

有的人会被**物质**刺激，
有的人会被**颜值**刺激，
有的人会被**精神**刺激
还有的人是**佛系**的

找到对方的共振频率，
才有可能找到共鸣

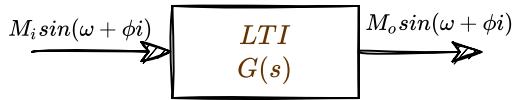
"频率的作用":
和有的人相处舒服
和有的人相处尴尬



动态系统的建模与分析

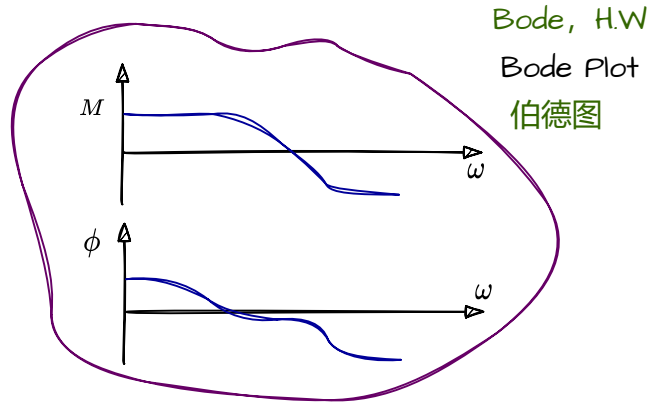
Bode Plot 伯德图 (i)

对于一个线性时不变系统



振幅的变化 $M = \frac{M_o}{M_i} = |G(j\omega)|$

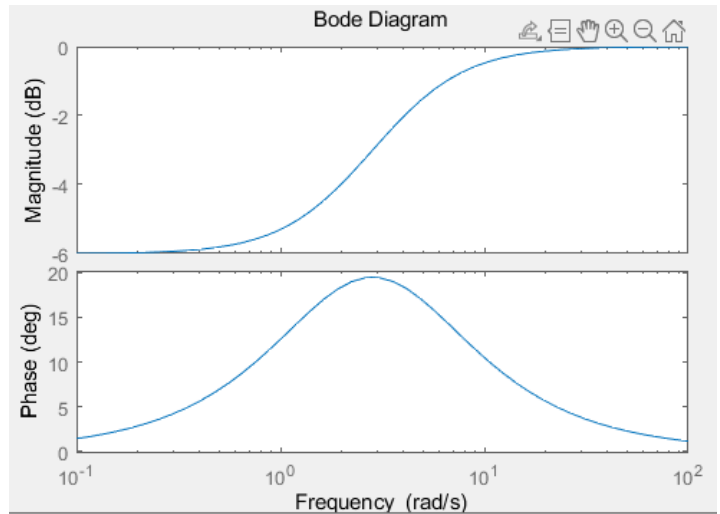
相位的变化 $\phi = \phi_o - \phi_i = \angle G(j\omega)$



matlab bode()

$$G(s) = \frac{s+2}{s+4}$$

`>>bode([1 2],[1 4])`



dB decibel 分贝
alexander bell
 $\frac{1}{10}$
dm

60 dB 交流、交谈
80dB 闹市



强度差 100倍

电话电报的信号损失

$$dB = 10 \log_{10} \frac{P_M}{P_R}$$

Measurement 测量
Reference 参考

能量的比值取对数

取对数的原因是把一个很大数缩小

声音强度

可听见的最低音量
电锯

$$1 \times 10^{-12} \text{ W/m}^2$$

$$1 \text{ W/m}^2$$

$$10 \log_{10} 1 = 0 \text{ dB}$$

$$10 \log_{10} 10^{12} = 120 \text{ dB}$$

在bode图中 $20 \log_{10}$

$$M = \frac{M_o}{M_i}$$

一般来说, 振幅和功率是平方的关系

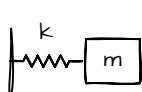
功率
能量

$$P = f(M^2)$$

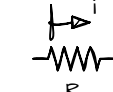
振幅



$$\frac{1}{2}mv^2$$



$$\frac{1}{2}kx^2$$



$$i^2 R$$

$$dB = 10 \log_{10} \frac{P_M}{P_R} = 10 \log_{10} \left(\frac{M_o}{M_i} \right)^2 = 20 \log_{10} M$$

积分 Integrator

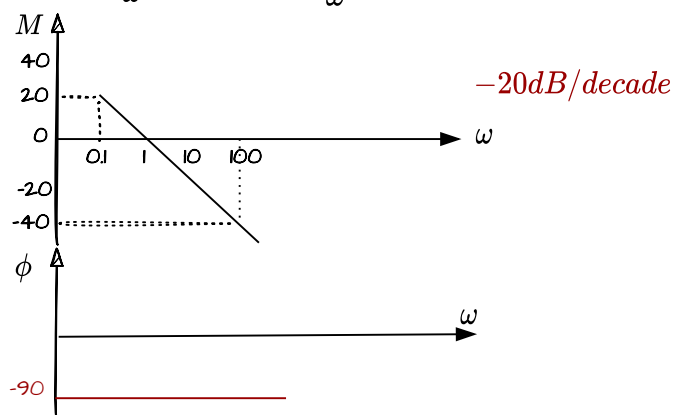
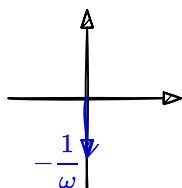
$$G(s) = \frac{1}{s}$$

$$G(j\omega) = \frac{1}{j\omega} = -\frac{1}{\omega}j$$

$$|G(j\omega)| = \frac{1}{\omega}$$

$$20 \log \frac{1}{\omega} = 20 \log \omega^{-1} = -20 \log \omega$$

复平面



$$G(s) = \frac{a}{s+a}$$

注: 推导见一阶系统的频率响应

$$|G(j\omega)| = \sqrt{\frac{1}{1 + \left(\frac{\omega}{a}\right)^2}}$$

$$\angle G(j\omega) = -\arctan\left(\frac{\omega}{a}\right)$$

低频 $\omega \ll a$

$$|G(j\omega)| = 1$$

$$20 \log |G(j\omega)| = 0$$

$$\angle G(j\omega) = 0$$

这个 $-3dB$, 输入的振幅是输出的振幅的 $\sqrt{\frac{1}{2}}$, 从能量的角度是输出的能量是输入的 $\frac{1}{2}$

截至频率 $\omega = a$

$$|G(j\omega)| = \sqrt{\frac{1}{2}}$$

$$20 \log G(j\omega) = 20 \log (\sqrt{2})^{-1} = -20 \log \sqrt{2} = -3dB$$

$$\angle G(j\omega) = -\arctan 1 = -45^\circ$$

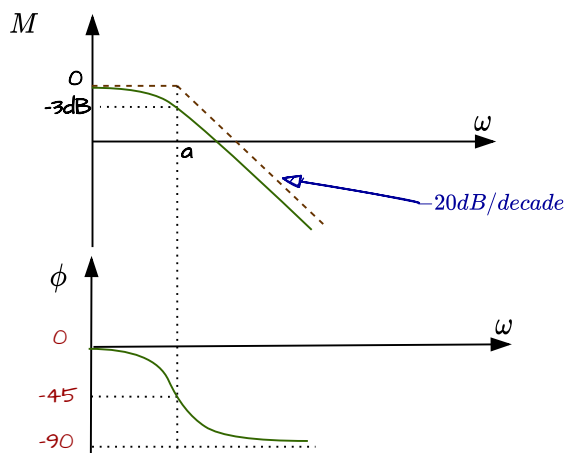
高频 $\omega \gg a$

$$|G(j\omega)| = \frac{1}{\omega}$$

$$20 \log \frac{1}{\omega} = 20 \log \omega^{-1} = -20 \log \omega$$

$-20dB/decade$

$$\angle G(j\omega) = -90^\circ$$



试一试

$$G(s) = s$$

$$G'(s) = as + a$$

bode图的另一个好处:

把乘积变成加和

$$\log_{10} AB = \log_{10} A + \log_{10} B$$

把复杂的传递函数分解

动态系统的建模与分析

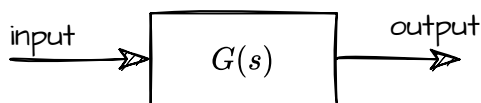
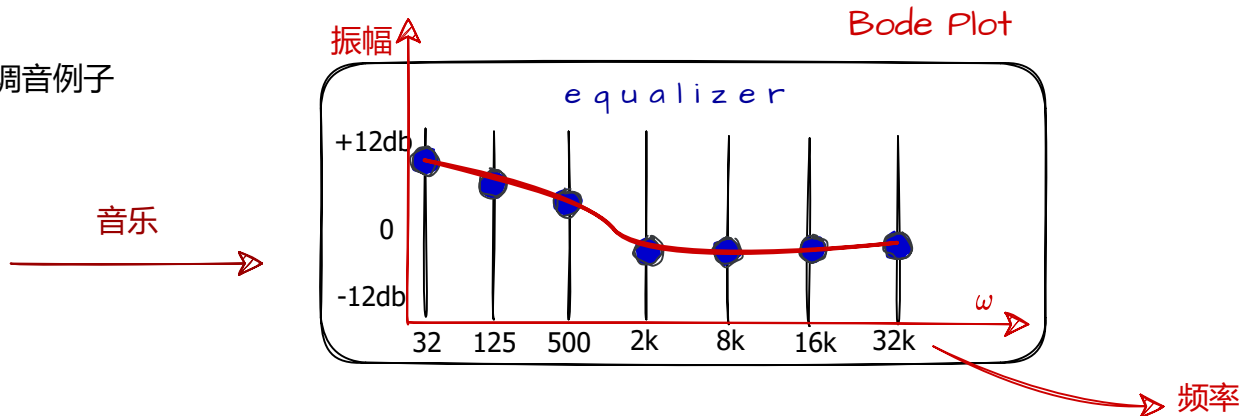
Bode Plot 伯德图 (2)

Year of 9102

数学规则

应用

调音例子



振幅的变化 $M = \frac{M_o}{M_i} = |G(j\omega)|$

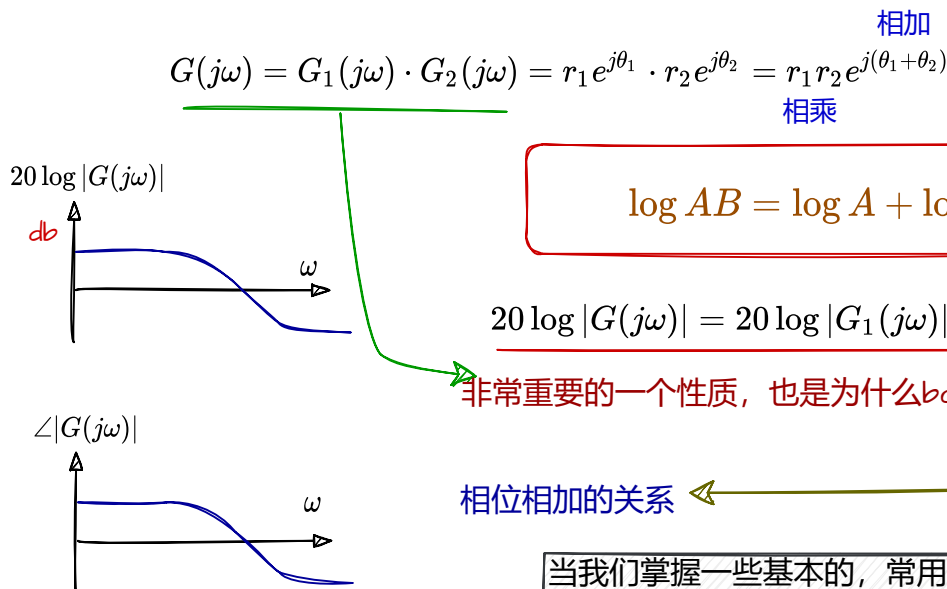
复数

$$G(j\omega) = re^{j\theta}$$

$$r = |G(j\omega)|$$

相位的变化 $\phi = \phi_o - \phi_i = \angle G(j\omega)$

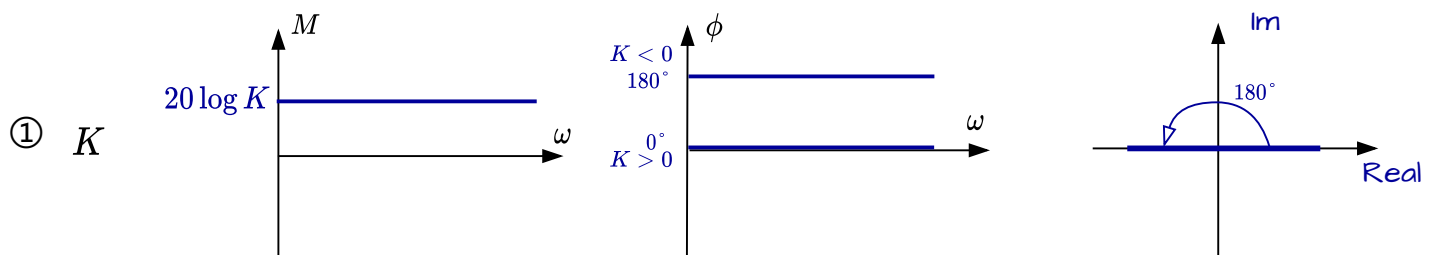
$$\theta = \angle G(j\omega)$$

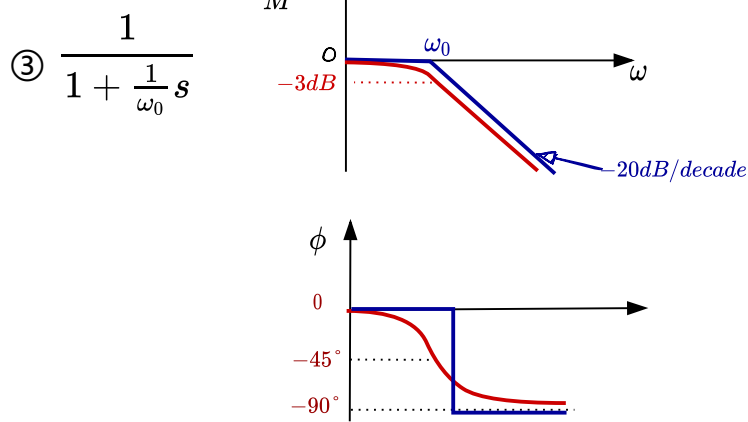
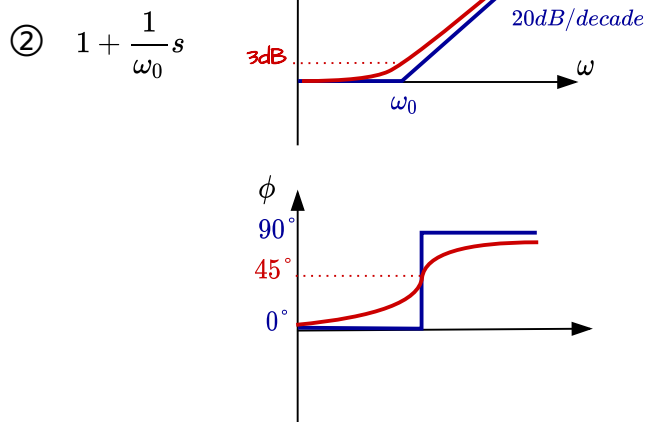


相位相加的关系

当我们掌握一些基本的, 常用的传递函数的bode的绘制
对于复杂系统来说, 只是**拆分组合**

常用:





上面三个作为基础，就可以做组合了

例 $G(s) = \frac{s+4}{s+8} = \frac{4(\frac{1}{4}s+1)}{8(\frac{1}{8}s+1)} = \frac{1}{2} \cdot (\frac{1}{4}s+1) \frac{1}{\frac{1}{8}s+1}$

① ② ③

